CTIS 256

Final – Duration 120 Mins

Setup & Hints

- Log into "phpmyadmin", http://localhost/phpmyadmin
- Username: "**std**" Password: "", no password, **emtpy string**.
- Import "**items.sql**" file into "**test**" database
- There is only one table: **items**(id, name, value, weight, image)
    - Name: item name
    - Value: value or price of the item ($)
    - Weight: weight of the item (kg)
    - Image: the image of the item under images folder.
- ***Hints about SQL Queries:***
    - *select * from items order by id*
    - *You can assume that the IDs of the items in the table start from 1 and increase by 1.*
- Watch "**demo.mp4**" to find out what you are supposed to do.
- You are not allowed to update/change the given table (items).
- HTML Codes: ✕ : **✕**

OBJECTIVE

In this application, you're given a list of items, each with a specific weight and value. Your task is to place these items into a backpack with a capacity of **20 kg**. You can select items from the left panel and add them to the backpack. When you add an item, its weight is subtracted from the backpack's remaining capacity, and the total value of the items in the backpack is displayed.

If you wish, you can also remove items from the backpack, which updates the backpack's status accordingly. You can only add items that fit within the remaining capacity. For example, if there are 5 kg left in the backpack, you won't be allowed to add an item that weighs 6 kg, Fig 1.

By clicking the 'Reset' link, you can empty the backpack. This creates a user interface that allows users to experiment and try to fill the backpack with the highest-value combination of items.

At the top, the interface keeps track of the highest total value you've achieved so far. Clicking on it shows you the items that made up that value.

Attention: To store the content of the bag (backpack), **you have to use "php session" to simulate the bag**. You can store ids of items in the session like $_SESSION["bag"] which is initially an empty array []. "Add" link of an item pushes the related item's id to this session array.

REQUIREMENTS

a. (**10Pts**) The approximate page design (layout, spacing, color). Font family is 'Segoe UI'.

b. (**20Pts**) It shows the items (fruits) on the left panel. Image width is 30px.

c. (**5Pts**) "Add" link adds the item to the bag, it does not show on the items panel any more.

d. (**15Pts**) It shows the added item on the right panel as part of the bag. It shows the item's image, its weight, value and an icon to remove from the bag. Image width is 50px.

e. (**10Pts**) It shows the items that fit in the bag with Add button, otherwise, show them light gray without Add button. (if the remaining capacity of the bag is 5 kg, you cannot add an item with 6 kg)

f. (**5Pts**) It keeps the status of the bag up-to-date (Remaining and Value parts in red)

g. (**5Pts**) “Reset” button removes all the items in the bag (session).

h. (**10Pts**) “╳” icon of an item in the bag removes the item from the bag.

i. (**5Pts**) “Maximum Value so far” shows the maximum value of the bag up to now. You can store this value, and the items in session.

j. **(15Pts)** If you click on “Maximum value so far”, it sends a GET request to a web service to retrieve the items that resulted in that value, It shows the images of the items at the bottom of the page. It shows a spinning icon (*spinner.gif*) when it starts AJAX, and removes as it gets the result. The web service <u>does not</u> have to be RESTfull compatible. Use 1 second delay to slow down the execution of the web service just to show spinning icon.

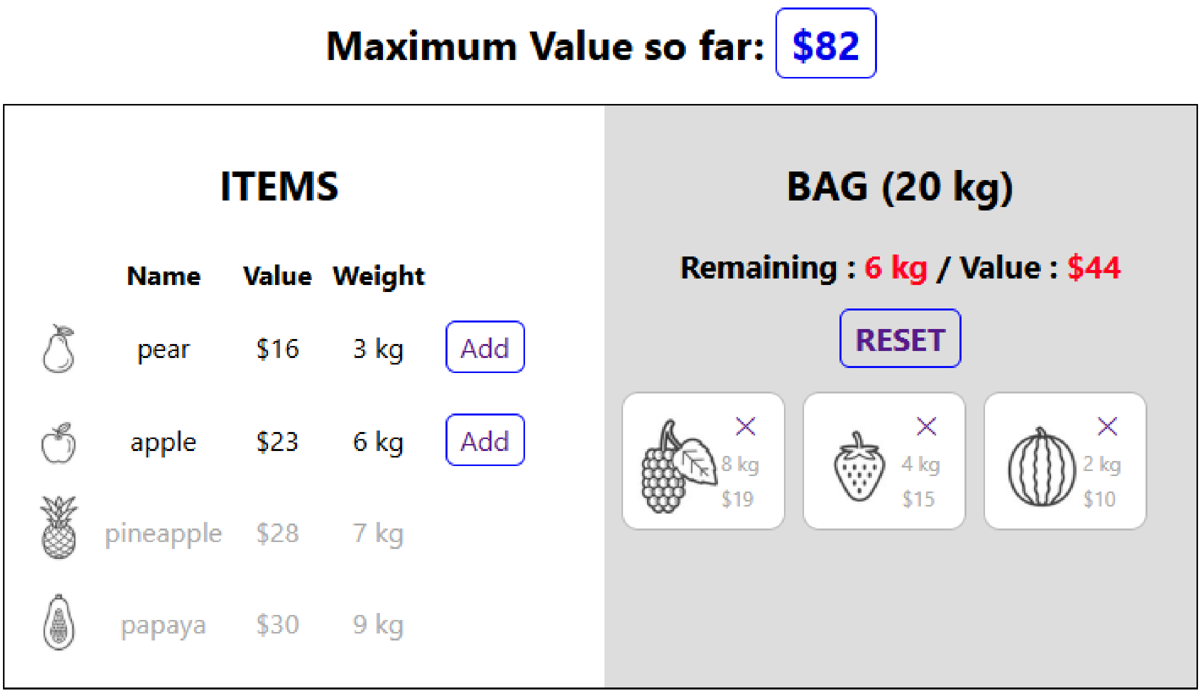

Fig1: Bag Status

Attention: Partial grade is applicable only for running codes.

- Good Luck -